

ENBC321: Homework #1

Due: Monday, 02/09/26 by 5:00 PM, beginning of the class

Submit your homework to ELMS, including Python code and the dataset

Overall points: 100

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.py". Start with the problem number and followed by your full name, e.g., "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of ZERO. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- Python code files
- Your answers to the questions in PDF format

Linear Algebra Concepts in Python

In this assignment, you will use **Python** and the **NumPy library** to explore key linear algebra concepts.

- You are required to clearly write out your answers by hand and provide the corresponding Python code.

Problem 1. Solve the following problems analytically (by hand) and computationally using Python. (40 points)

Given the matrix and vector:

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 7 \\ 4 \end{bmatrix}$$

- a) Solving a Linear System. Given the system of equations: $Ax = b$, solve for x and verify the solution. (4 points)
- b) Compute the **determinant** of A and explain its significance in solving linear systems. (4 points)
- c) Compute the **transpose** of A . (4 points)
- d) Compute the **trace** of A . (4 points)
- e) Compute the **rank** of A . (4 points)
- f) Check if the columns of A are **linearly independent**. (4 points)
- g) Verify that A is **orthogonal** $A^T A = I$. (4 points)
- h) Compute **inverse** of A and confirm $A^{-1} = A^T$. (4 points)
- i) Compute the **eigenvalues** and **eigenvectors** of A . (4 points)
- j) Compute the **gradient** and **Hessian matrix** of the function: (4 points)

$$f(x, y, z) = x^2 + 3xy + y^2 + 2xz + z^2$$

Problem 2. Solve the following problems analytically (by hand) and computationally using Python. (20 points)

Given the matrix and vectors:

$$M = \begin{bmatrix} 14 & -7 & -10 \\ 2 & 9 & 5 \\ 14 & -22 & 16 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 2 \\ 8 \\ 6 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 13 \\ 21 \\ 16 \end{bmatrix}$$

- a) Compute the **dot product** $v_1^T v_2$. (4 points)
- b) Compute the **matrix-vector product** $M v_1$. (4 points)
- c) Compute the **matrix-matrix product** $M^T M$. (4 points)
- d) Check if M is **symmetric** and explain why. (4 points)
- e) Compute the **L2 norm**, **L1 norm**, and **L $^\infty$ norm** of v_1 and v_2 . (4 points)

Problem 3. In this problem, you will explore different data cleaning techniques using Python and apply them to a real-world dataset. (40 points)

Find a dataset from **Kaggle** or any other online open dataset repository. Ensure that your dataset contains missing values and duplicates. Apply the following data cleaning techniques step by step and document the changes at each stage. After applying each technique, compute the **mean**, **variance**, and **standard deviation** of **ONLY ONE** numeric column.

- a) **Remove rows** with missing values and observe how much data is lost. Compute and compare the mean, variance, and standard deviation before and after removal. (7 points)
- b) Fill missing values in numeric columns using **mean**. Compute and compare the mean, variance, and standard deviation before and after removal. (6 points)
- c) Fill missing values in numeric columns using **median**. Compute and compare the mean, variance, and standard deviation before and after removal. (6 points)
- d) Use **forward fill** (`ffill`) and observe its effect. Compute and compare the mean, variance, and standard deviation before and after removal. (7 points)
- e) Use **backward fill** (`bfill`) and compare the differences. Compute and compare the mean, variance, and standard deviation before and after removal. (7 points)
- f) Identify and **remove duplicate rows** from the dataset if any. Compute and compare the mean, variance, and standard deviation before and after removal. (7 points)